

NAS is a file level storage system designed for serving files. It uses the standard network and file sharing protocols, such as FTP, SMB and CIFS. Unlike DAS, SAN operates on blocks. It uses IP and fibre channels to connect with servers. Figures 1, 2 and 3 show the architecture of DAS, NAS and SAN storage systems, respectively.

Benefits of centralised storage

There are several benefits of moving away from DAS to a centralised storage system. A few important ones to consider are:

- a) Centralised storage improves the utilisation of disks. Thin provisioning makes it appear to the servers that they have more storage than has actually been paid for. Disk utilisation of centralised storage can be further improved by space-saving technologies such as deduplication and compression. Deduplication detects and eliminates identical blocks, and the eliminated blocks are replaced by a pointer to a single copy of the block on the disk. Compression detects and eliminates redundant data and white spaces in files.
- b) Storage costs are reduced because of cost savings on hardware, rack space, power and cooling.
- c) Centralised storage supports technologies like RAID (redundant array of inexpensive disks), which help to stripe data across multiple disks in enterprise class storage systems, improving performance and capacity for data in disks. NAS and SAN storage systems are built for resiliency. If a disk fails, redundant components can take its place.
- d) It is much easier to manage storage from a centralised location than separately across servers.
- e) SAN protocols allow clients to

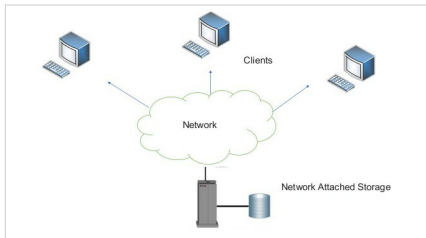


Figure 2: Network attached storage

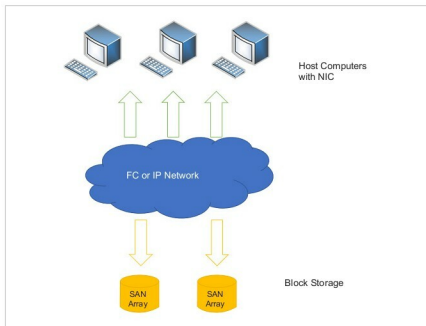


Figure 3: Storage area network

- boot from logical disks on remote storage and, therefore, the clients, such as blade servers, do not need to contain any disks.
- f) Centralised storage systems allow for storage tiering, which analyses and stores data on media with different attributes. For example, 'hot' data is stored in high-performance SSDs, and 'cold' data gets archived into high-capacity and low-performance SATA drives.
- g) Centralised storage provides centralised backup solutions.
- h) It allows snapshots, for convenient backup and restoration.
- i) It also allows replicating data to a disaster recovery site.
- j) Virtualisation support and vMotion: Software such as VMware and Hyper-V allow multiple virtual servers to run in the same underlying physical server. Virtual servers can be moved between physical servers on the fly.